
MHF 4U1

CHAPTER 8: COMBINING FUNCTIONS

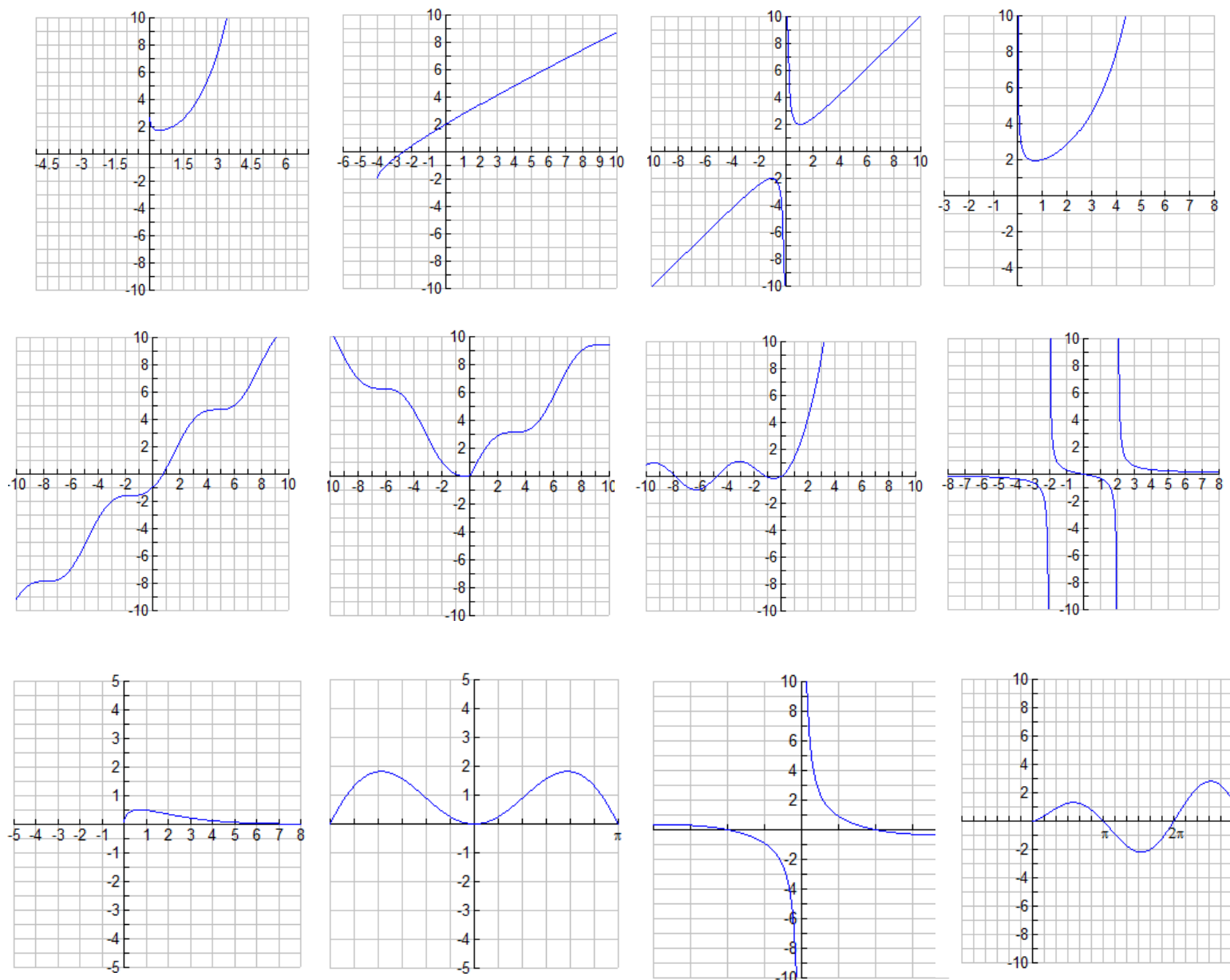
Day	Section	Homework	Completion
1	8.1 Sums and Differences of Functions	Page 424 # (3-5)eo,6-8,10,14,16	
1	8.2 Product and Quotient of Functions	Page 435 # 1,2,4,17(with table) 18	
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1	8.4 Inequalities of Combined Functions	Page 457 # 3, 5, 8, 14, 16	
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1	CHAPTER 8 TEST		

CHAPTER 8 TEST DATE:

Within this course, we have examined the behavior and discussed the properties of a variety of functions; absolute value and square root functions, polynomial functions, rational function, exponential and logarithmic functions, and trigonometric functions.

In this chapter, we will look at new ways of combining these functions to create more complex functions. This can be done through **addition, subtraction, multiplication and division** of two or more simpler functions.

Given the graphs pictured below. They were created by combining two familiar functions.



Compare each of the graphs above with the function equations in the table below. **Can you predict which equations will match each graph?** Match the following equations with the graphs above and verify using desmos.

$y = \frac{1}{2}x + \sqrt{x+4}$	$y = x + \frac{1}{x}$	$y = 2^x - \cos x$	$y = x \sin x$	$y = 2^x - \log(x)$	$y = x - \cos x$
$y = \frac{x}{x^2 - 4}$	$y = \frac{1}{x} \cos x$	$y = x + \sin x$	$y = \frac{2^x}{\sqrt{x}}$	$y = \frac{\sqrt{x}}{2^x}$	$y = \sqrt{x}(\sin x)$

8.1 SUM OR DIFFERENCE OF FUNCTIONS

- Some combined functions are formed by adding or subtracting two or more functions.
- The **superposition principle** states that the sum of two or more functions can be found by adding the y-coordinate at every x value.

Ex. 1: Let $f = \{(-4, 4), (-2, 4), (1, 3), (3, 5), (4, 6)\}$
 $g = \{(-4, 2), (0, 2), (1, 2), (2, 2), (4, 4)\}$

The domains must be the same in order to add or subtract functions.

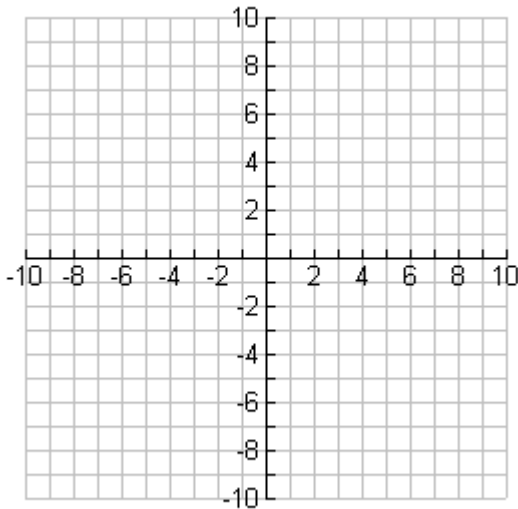
Determine:

a) $f + g =$

b) $f - g =$

c) $g - f =$

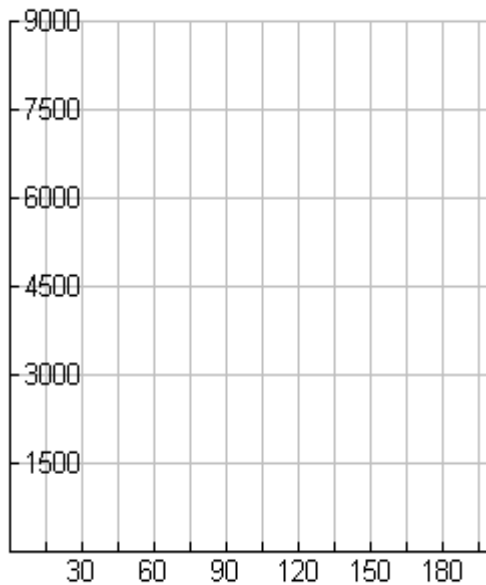
Ex. 2: Given $f(x) = -x^2 + 3$ and $g(x) = -2x$, determine an equation and draw the graph for the new function $y = f(x) + g(x)$. State the domain and range of the new function.



Ex. 3: Let $f(x) = x^2 - nx + 5$ and $g(x) = mx^2 + x - 3$. The functions are combined to form the new function $h(x) = f(x) + g(x)$. Points $(1, 3)$ and $(-2, 18)$ satisfy the new function. Determine the values of m and n . Write a simplified equation for $h(x)$.

Ex. 4: Student Council is selling prom tickets. The cost to rent the hall is \$1500, plus a meal cost of \$45 per ticket. The council has decided to sell the tickets for \$60 each.

- a) Write an equation to represent
- The total cost, C , as a function of the number, x , of prom ticket sold.
 - The revenue, R , as a function of the number, x , of prom tickets sold.
- b) Graph the functions in part a) on the same set of axes. Identify the point of intersection and explain the meaning of its coordinates.



- c) Under what circumstances will Student Council lose money? Make money?
- d) Profit, P , is the difference between revenue and expenses. Develop an algebraic model for the profit function. Include the function $P(x)$ on the above grid.
- e) Identify the domain and range of the cost, revenue, and profit functions in the context of this problem.

Try it now!

1. If $f(x) = ax^2 + ax - 1$, $g(x) = bx^2 + 3x - 5$ and $(f - g)(x) = 4x^2 - ax + c$, determine the values of a , b , c .

2. Consider $f(x)$ and $g(x)$ each defined by one of the equations below:

$$y = \sin x$$

$$y = \cos x$$

$$y = x$$

$$y = x^2$$

$$y = x^3$$

$$y = 2^x$$

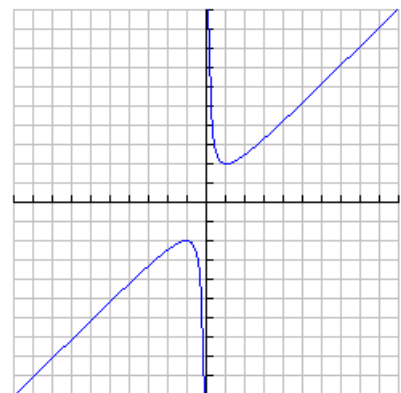
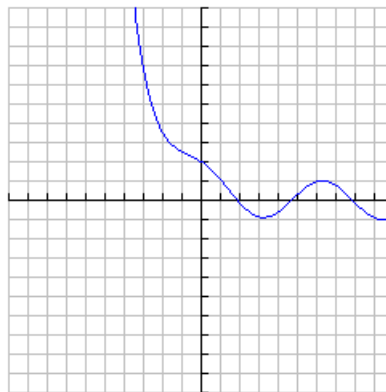
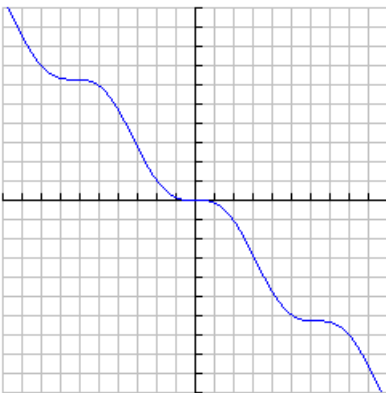
$$y = \frac{1}{x}$$

$$y = |x|$$

$$y = \sqrt{x}$$

$$y = \left(\frac{1}{2}\right)^x$$

Determine the sum ($y = f(x) + g(x)$) or difference ($y = f(x) - g(x)$) or ($y = g(x) - f(x)$) that was used to graph each of the following:



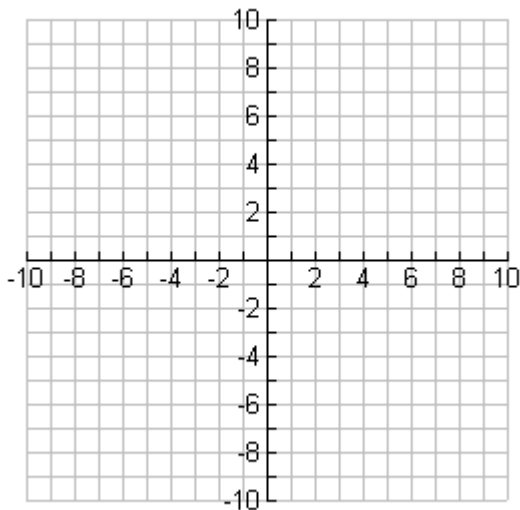
8.2 PRODUCT AND QUOTIENT OF FUNCTIONS

A combined function of the form $y = f(x)g(x)$ represents the product of two functions, $f(x)$ and $g(x)$.

The combined function $y = \frac{f(x)}{g(x)}$ represents the quotient of the two functions, $f(x)$ and $g(x)$, for $g(x) \neq 0$.

The domain of the product and quotient of functions is the domain **common** to the component functions. The domain of a quotient function, $y = \frac{f(x)}{g(x)}$, is further restricted by excluding any values that make the denominator, $g(x)$, equal to zero.

Ex. 1: Given $f(x) = x^2 + x - 6$ and $g(x) = x$. Determine the graph of $y = f(x)g(x)$ by graphing $f(x)$ and $g(x)$.



Ex. 2: Let $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{2}x - 2$.

a) Find the equation of $y = f(x)g(x)$.

c) Find the domain of $y = f(x)g(x)$.

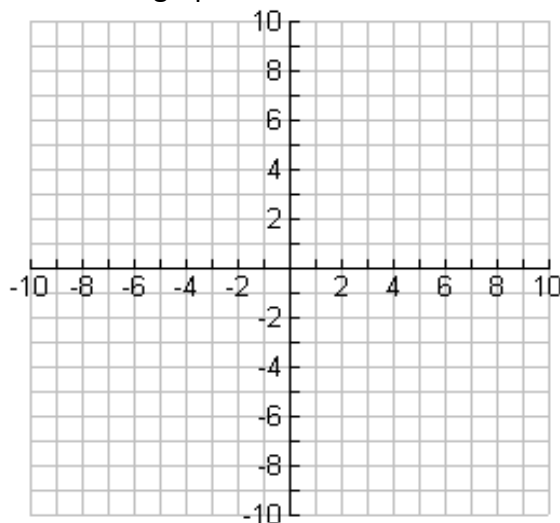
b) Determine $y = f(4)g(4)$.

Recall: Even function: $f(-x) = f(x)$ --> symmetric about the y-axis
(line symmetry)

Odd function: $f(-x) = -f(x)$ --> symmetric about the origin
(point symmetry)

Ex. 3: Let $f(x) = 2x + 3$ and $g(x) = 2x^2 + x - 3$. Determine an equation for $y = \frac{f(x)}{g(x)}$.

- a) Sketch a graph of the combined function and state its domain and range.



- b) Is y an even function, odd function or neither?

Ex 4: Let $f(x) = 5x^2 - 20$ and $g(x) = x^3 - 4x$. Solve for $y = \frac{f(x)}{g(x)}$. Is y even, odd or neither?

Summary for the product & quotient of even/odd functions:

8.3 COMPOSITE FUNCTIONS

A **composite function** is a function that depends on another function. It is formed when one function is substituted into another.

read "f operates on g"

read "f of g of x"

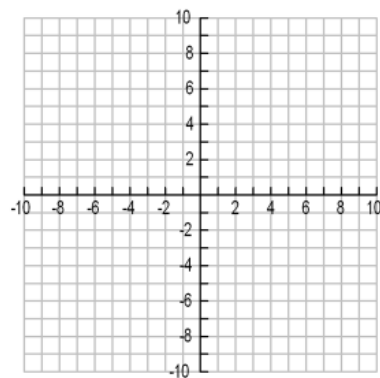
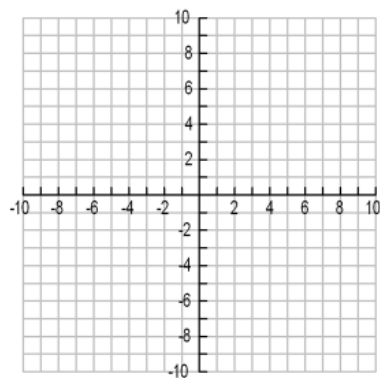
$$(f \circ g)(x) = f(g(x))$$

This is the composite function that is formed when $g(x)$ is substituted for x in $f(x)$.

Ex.1: Given the functions $f(x) = 2x + 3$ and $g(x) = \sqrt{x}$, determine an equation for each composite function, graph the new function and give its domain and range.

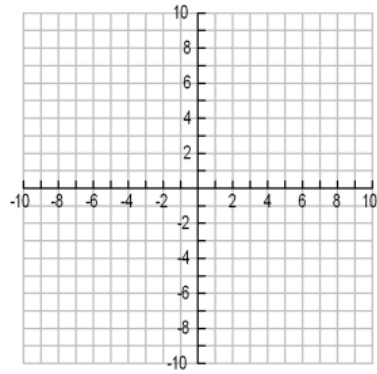
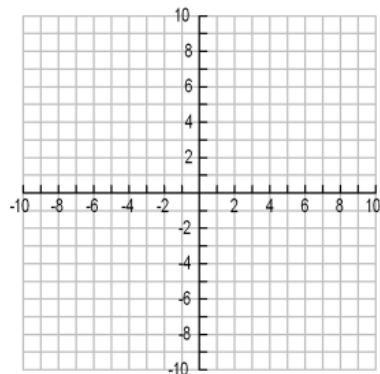
a) $y = f(g(x))$

b) $y = g(f(x))$



c) $y = f(f^{-1}(x))$

d) $y = f^{-1}(f(x))$



Ex.2: If $f(x) = \frac{1}{x}$ and $g(x) = 4 - x$, determine each of the following, if it exists.

a) $f(g(3))$

b) $f(g(0))$

c) $f(g(4))$

d) $g(f(4))$

e) $f\left(f^{-1}\left(\frac{63}{58}\right)\right)$

f) $g^{-1}\left(g\left(3k^2 - 2k\right)\right)$

g) $f^{-1}(g(3x - 1))$

h) $g^{-1}(f^{-1}(25))$

Try it now!

1. Given the table of values below, find the following:

x	$f(x)$	$g(x)$
1	5	6
2	3	5
3	1	4
4	3	3
5	5	2

a) $f(g(2))$

b) $(f \circ f)(4)$

c) $f(g(1))$

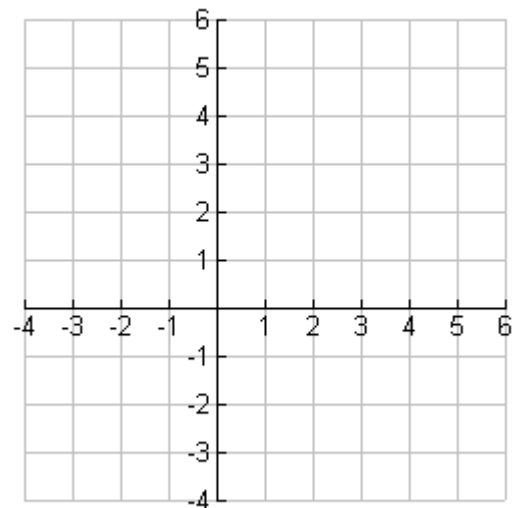
d) State the domain of $(f \circ g)(x)$ _____

2. Given $f(x) = 4 - x^2$ and $g(x) = \sqrt{x}$.

a) State the domain of $f(x)$ _____

State the domain of $g(x)$ _____

b) Find the equations of $y = (f \circ g)(x)$ and $y = (g \circ f)(x)$



c) Sketch the graphs of $y = (f \circ g)(x)$ and $y = (g \circ f)(x)$.

3. Given $f(x) = \left(\frac{1}{3}\right)^{1-x}$, $g(x) = 1 - \log_3(x)$, $h(x) = \sqrt{2-x}$.

a) Evaluate $h(g(1))$ b) Find $(g \circ f)(x)$ in simplest form and state the domain

c) Find $(g \circ h)(x)$ in simplest form and state the domain

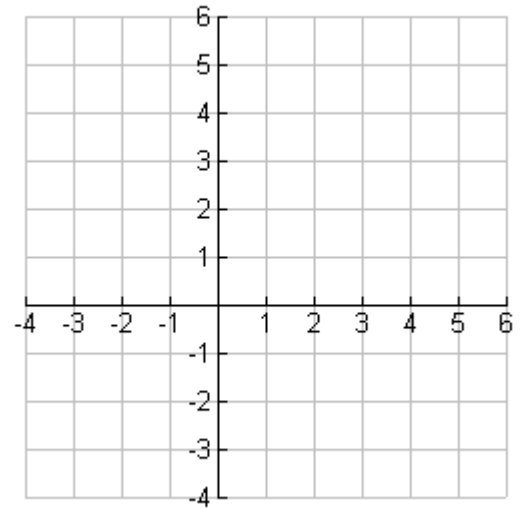
Try it now!

1. Given $f(x) = 9 - x^2$ and $g(x) = \sqrt{x}$.

a) Find the equations of $y = (f \circ g)(x)$ and $y = (g \circ f)(x)$

b) Sketch the graphs of $y = (f \circ g)(x)$ and $y = (g \circ f)(x)$.

c) State the range of each.



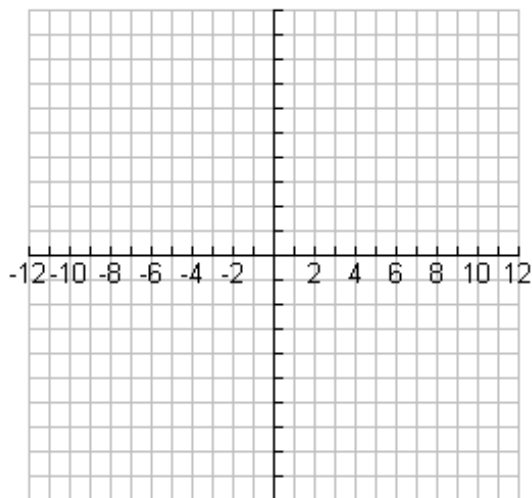
2. TRUE OR FALSE

a) When two functions are added, the domain of the combined function consists of all of the values common to the domain of both of the original functions.

b) If $f(x)$ is a function that is defined for all $x \in \mathbb{R}$, then $f(f^{-1}(x)) = x$.

c) When two functions are divided, the domain of the combined function consists of all of the values in the domains of the original functions.

3. Let $f(x) = -2x+1$, $g(x) = x^2-2x+1$, and $h(x) = x^3-1$. Determine graphically all values of x such that $f(x) < g(x) < h(x)$.



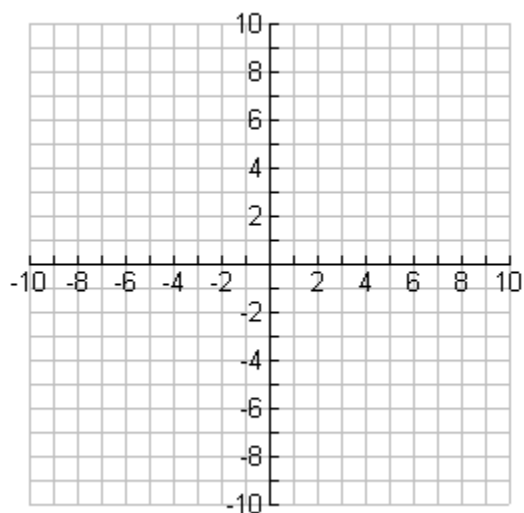
8.4 INEQUALITIES OF COMBINED FUNCTIONS

Solutions to problems involving combined functions can sometimes lead to a range of acceptable answers. When this happens, techniques for solving inequalities are applied.

Ex#1: Let $f(x) = |x|$ and $g(x) = (x - 2)^2$.

a) Graph both functions below. Identify the points of intersection.

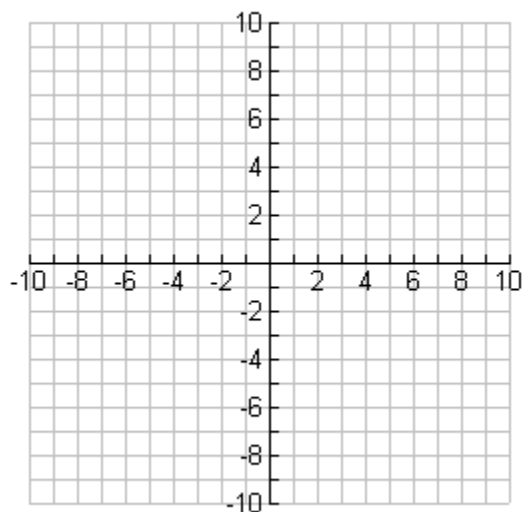
b) State the region for which: i) $f(x) > g(x)$ and ii) $g(x) > f(x)$.



Ex#2: Let $f(x) = x^2 - 3$ and $g(x) = 2x$.

a) Graph both functions below. Identify the points of intersection.

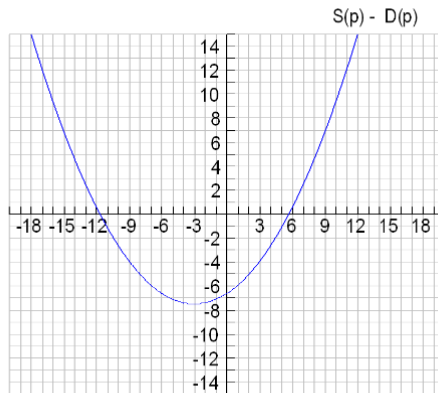
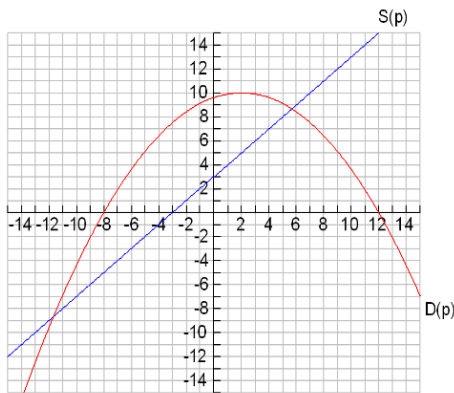
b) State the region for which: i) $f(x) > g(x)$ and ii) $g(x) > f(x)$.



Page 458, Challenge Question

- Hint for #5:
- i) first graph the quadratic do not graph the cubic (you need a graphing calculator)
 - ii) solve algebraically for the intersection points: from the Factor Theorem $f(2) = 0$, then use the quadratic formula to find the other two intersection points
 - iii) sketch the cubic based on the three intersection points and end behaviour
 - iv) solve the inequality

Page 473 Chapter Problem solutions to parts c) a) b) d):



- c) In the context of this situation, this graph is only defined for $p > 0$; as p represents the price of the teddy bear. There is only one valid POI at $(5.66, 8.66)$. The other POI located at $(-11.66, -8.66)$ is extraneous. The coordinates of the valid POI mean that when the price is set at \$5.66, the supply of teddy bears will equal the demand of teddy bears, 8660.

- a) $D(p) > S(p)$ when $-11.66 < p < 5.66$, but again, this makes no sense for negative values of p , so:

$$D(p) > S(p) \text{ when } 0 < p < 5.66$$

This implies that the demand for teddy bears is higher than the supply of teddy bears; everybody wants to buy them and the price goes up!

- b) $D(p) < S(p)$ when $5.66 < p < 12$.

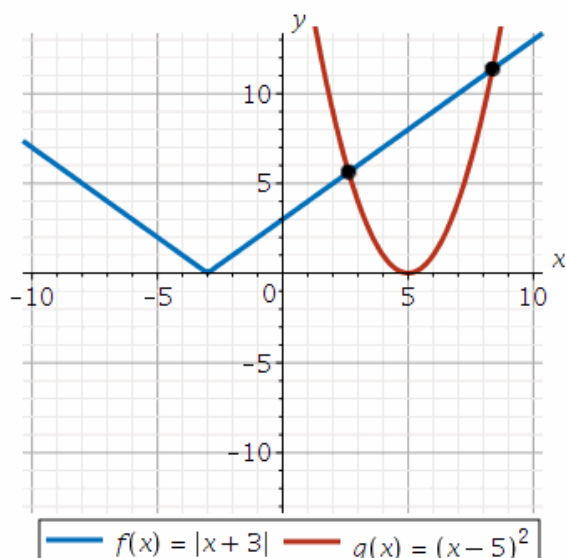
This implies that the supply for teddy bears is higher than the demand of teddy bears; nobody wants to buy them and the price goes down!

- d) This graph shows there is too much supply when the graph is positive (above the x-axis) and not enough supply (too much demand) when the graph is negative (below the x-axis).

Try it now!

Given the graphs and the coordinates of all points of intersection of the two functions, $f(x)=|x+3|$ and $g(x)=(x-5)^2$, solve $|x+3| > (x-5)^2$.

The points of intersection of the two functions are $(2.628, 5.628)$ and $(8.372, 11.372)$, accurate to 3 decimal places.

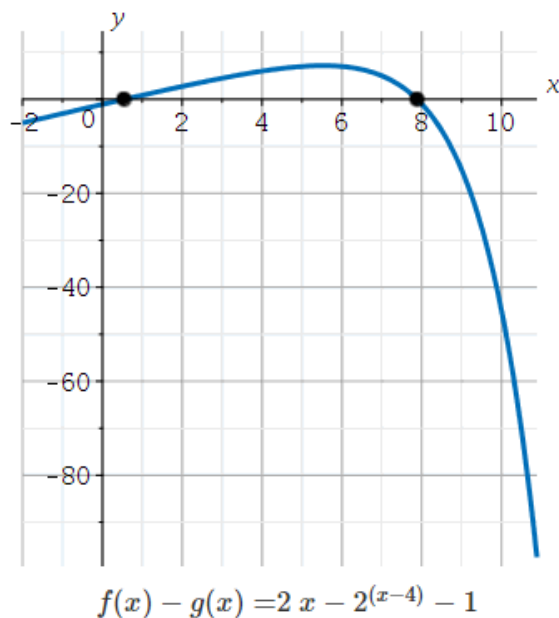


Using the checkboxes, select **all** appropriate interval(s) which are part of the solution.

- ☐ $x < -6$
- ☐ $-6 < x < -3$
- ☐ $-3 < x < 2.628$
- ☐ $2.628 < x < 5.5$
- ☐ $5.5 < x < 8.372$
- ☐ $x > 8.372$

Given the graph of $y = f(x) - g(x)$, where $f(x) = 2x$ and $g(x) = 2^{(x-4)} + 1$, solve $2x - 2^{(x-4)} - 1 \leq 0$.

The x -intercepts of $y = f(x) - g(x)$ are $(0.546, 0)$ and $(7.885, 0)$, accurate to 3 decimal places.



Using the checkboxes, select **all** appropriate interval(s) which are part of the solution.

- ☐ $x < -1$
- ☐ $-1 \leq x < 0$
- ☐ $0 \leq x \leq 0.546$
- ☐ $0.546 < x < 4.216$
- ☐ $4.216 \leq x < 7.885$
- ☐ $x \geq 7.885$

1. If $f = \{(-11, 7), (-7, -6), (-2, 16), (1, 3), (8, 13), (9, 4)\}$ and $g = \{(-11, -2), (-8, 5), (-7, 3), (-2, 1), (2, 12), (8, 6)\}$ evaluate each of the following expressions

a) $f + g$ b) $f - g$ c) $g + f$ d) $g - f$ e) $f - f$ f) $g + g$

2. For each of the following pairs of functions, determine $(f \times g)(x)$.

a) $f(x) = 4x - 1$; $g(x) = x + 9$ b) $f(x) = \sqrt{32x^3}$; $g(x) = \sqrt{2x^5}$
 c) $f(x) = 9\sin x$; $g(x) = 3\sec x$ d) $f(x) = \frac{x^2}{4}$; $g(x) = \frac{96}{x}$

3. For each of the following functions, determine $(f \div g)(x)$.

a) $f(x) = 6x^2 - 26x + 24$, $g(x) = 2x - 6$ b) $f(x) = 21\cos x$; $g(x) = 3\sin x$
 c) $f(x) = 7x - 8$; $g(x) = \sqrt{7x - 8}$ d) $f(x) = 25^x$, $g(x) = 5^x$

4. State the domain of $(f \div g)(x)$ where $f(x) = 11x^5$ and $g(x) = 22x^2 \log x$.

5. If $f(x) = 6x + 1$ and $g(x) = \frac{3x^2}{4}$, evaluate each of the following expressions.

a) $f(g(-3))$ b) $g(f(4))$ c) $(f \circ g)\left(\frac{1}{4}\right)$ d) $(g \circ f)(0)$
 e) $(f \circ f^{-1})\left(-\frac{3}{4}\right)$ f) $(g \circ f^{-1})(5)$

6. Let $f(x) = 2x$ and $g(x) = x^2 - 1$.

- a) Graph the functions on the same set of axes. Identify the points of intersection.

- b) Find the regions for which:

(i) $f(x) > g(x)$ (ii) $g(x) > f(x)$

7. Solve $\sin x < x$.

8. Let $f(x) = \frac{1}{\sqrt{x+1}}$ and $g(x) = x^2 + 3$.

- a) What are the domain and range of $f(x)$ and $g(x)$?

- b) Find $f(g(x))$.

- c) Find $g(f(x))$.

- d) Find $f(g(0))$.

- e) Find $g(f(0))$.

- f) State the domain of each of the functions you found in parts b) and c)

Try it now!

1.a) Given $f(x) = 2\log_2(x + 15)$ and $g(x) = x^2$, determine $(f \circ g)(-1)$.

b) Give $f(x) = 2\log_2(x - 6)$ and $g(x) = \sqrt{x + 1}$, evaluate $(f \cdot g)(8)$

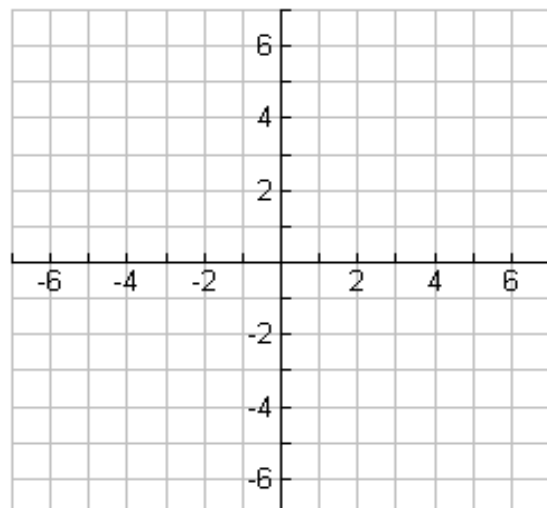
2. Given $f(x) = 3x + 2$ and $g(x) = 2^{x+3}$, identify the domain of $\left(\frac{f}{g}\right)(x)$. _____

3. If $f(x) = \tan x$ and $g(x) = \frac{3}{x}$, then is $(f \cdot g)(x)$ an even function, an odd function, or neither? _____

4. a) Determine, algebraically, all points of intersection of the two functions, $f(x) = -\frac{1}{2}x - 3$ and

$$g(x) = -2\sqrt{x + 2}.$$

b) Illustrate graphically.



c) Show on the graph where $-2\sqrt{x + 2} > -\frac{1}{2}x - 3$